

Bedanta Bandyopadhyay

Aspiring Electronics Engineer | ML Enthusiast | CAD Modeler | Product Visualizer

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SUMMARY

Results-oriented engineer merging a robust background in automotive CGI with emerging expertise in Deep Learning and IoT. Experienced in managing end-to-end client projects and delivering high-impact visual solutions for diverse brands. Adept at rapid skill acquisition and passionate about building intelligent, connected systems.

EDUCATION

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| 01. | Bachelor of Technology in Electronics and Communication Engineering
Institute of Engineering & Management, Kolkata | <i>Expected Graduation date: 2027</i>
<i>CGPA: 8.86</i> |
| 02. | Higher Secondary (ISC)
Ram Mohan Mission High School | <i>Percentage: 73.2%</i> |
| 03. | Secondary (ICSE)
Ram Mohan Mission High School | <i>Percentage: 94.5%</i> |

TECHNICAL SKILLS

TOOLS AND SOFTWARES:

General: Microsoft Office | Git

Circuit and Embedded: Cadence Virtuoso | LT Spice | MATLAB | HFSS | Scaps1D | Arduino | Raspberry Pi | Jetson

PCB Design: KiCAD

Artificial Intelligence: Scikit-Learn | LangChain | RAG | HuggingFace

Database Management: SQL | MongoDB

CAD Engineering: AutoCAD | SolidWorks | Fusion360

3D Modelling: Blender3D | Autodesk Maya | Autodesk 3DS Max

3D Rendering: Blender Cycles | Chaos Corona Renderer | Chaos V-Ray | Luxion Keyshot

Environment Design: Gaea

Postproduction and Retouching: Adobe Photoshop | Blackmagic DaVinci Resolve | Blackmagic Fusion Design | Foundry Nuke

2D Illustrations: Adobe Illustrator

VFX: Adobe Premier Pro | Adobe After Effects

Texturing: Adobe Substance Painter | Adobe Substance Designer

Cloud and API Architecture: AWS | Flask

Web Development: HTML | CSS | ReactJS | ExpressJS | NodeJS

PROGRAMMING LANGUAGES: Python | C/C++ | Java | Rust | JavaScript

CORE COMPETENCIES: Circuit Design | Circuit Simulation | Fluid Simulation | UI-UX | CAD Modeling | Database Management Systems | Signal Analysis | Electronic Troubleshooting

ACADEMIC PROJECTS

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| 01. | <u>OccuNova</u> (<i>Database Management Deep Learning</i>): Engineered Occunova, an end-to-end deep learning diagnostic system designed for automated glaucoma detection and optical nerve analytics. The solution features a robust computer vision architecture utilizing cascaded ResNet50 models integrated with advanced image processing techniques to analyze complex retinal imagery. This data is processed through AI-driven U-Net cropping of retinal regions in fundus scans, to optimize early medical intervention and is supported by a scalable inference backend. The project demonstrates full-lifecycle ML engineering, seamlessly bridging clinical data preprocessing, model training, and healthcare automation.
Tech Stack: Python, Deep Learning (CNNs), Computer Vision, AI/Data Analytics, Git/GitHub, NodeJS, ReactJS. | <i>Jun '26</i> |
| 02. | <u>Agro-Guard</u> (<i>Embedded Systems IoT Machine Learning</i>): Engineered AgroGuard, an end-to-end IoT agricultural system designed for real-time crop health monitoring and grain silo analytics. The solution features an embedded hardware architecture utilizing an Arduino microcontroller integrated with DHT11, HC-SR04, and LDR sensors to capture continuous environmental telemetry. This data is processed through AI-driven analytics to optimize storage management and is supported by a scalable backend deployed on Render. The project demonstrates full-lifecycle engineering, seamlessly bridging hardware integration, continuous data ingestion, and cloud deployment.
Tech Stack: Arduino, IoT Sensors (DHT11, HC-SR04, LDR), AI/Data Analytics, Render, Git/GitHub, NodeJS, ReactJS. | <i>Feb '26</i> |
| 03. | <u>Infrastructure-Assassin</u> (<i>Database Management Cloud LLM Cascade</i>): Engineered Infrastructure Assassin, an end-to-end AI-driven infrastructure testing suite designed for automated resilience evaluation and | <i>Jul '26</i> |

vulnerability identification. The solution features a multi-tier cloud architecture utilizing a PostgreSQL database for persistent telemetry logging and a Redis cache for low-latency session management, strictly governed by Role-Based Access Control (RBAC) to enforce secure operational execution. This complex infrastructure data is processed through a cascaded Large Language Model (LLM) architecture to optimize intelligent threat analysis and automated decision-making, supported by automated CI/CD pipelines and a scalable, multi-instance backend deployed on Render. The project demonstrates full-lifecycle engineering, seamlessly bridging stateful data management, advanced generative AI orchestration, and secure cloud deployment.

Tech Stack: Cascaded LLMs, PostgreSQL, Redis, RBAC, CI/CD, Cloud Infrastructure (Render), Git/GitHub, NodeJS, ReactJS.

PREVIOUS EXPERIENCES

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| 01. | Automotive and Product Visualizer: (<i>Freelance</i>): Extensive 7+ years tenure in remote freelance visualization. Specialized in hyper-accurate 3D design and lighting for automotive and product sectors. | <i>Apr '19 - Aug '25</i> |
| 02. | Co-Lead: (<i>Ninja Fellowship</i>): Led a team of 3d artists and worked alongside them to create striking visuals for various brands for over 2 years. | <i>Nov '22 - Mar '25</i> |
| 03. | 3D Generalist and Environment Artist: (<i>The AutoFocus Studio</i>): Specializing in environment design and generalist work. Delivered high-fidelity visualizations and technical artwork for studio projects and clients alike. | <i>Aug '25 - Feb '26</i> |

INTERNSHIPS

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| 01. | Internship on Advanced and Cleaner Materials in Solar Technology:
Conducted performance optimization of Tri-Layer Lead-free Carbon-based Perovskite Solar Cells using Machine Learning and crafted a full detailed report on the entire process. [PAPER] | <i>Jun '25 - Jul '25</i> |
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CERTIFICATIONS

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| 01. | Business Intelligence and Analytics: (<i>NPTEL, IIT Madras</i>): Teaches about various machine learning algorithms that are used by companies for prediction of consumer response and sales of products. [VIEW] | <i>Jun '25</i> |
| 02. | Joy of Computing with Python: (<i>NPTEL, IIT Ropar</i>): Teaches about the basics of python and goes up to the various advanced topics that form the foundation of data science and analysis. [VIEW] | <i>Nov '24</i> |
| 03. | Cloud Computing: (<i>NPTEL, IIT Kharagpur</i>): Teaches about Cloud Computing and how various aspects and types of services are used and availed by the firms. [VIEW] | <i>Jun '25</i> |
| 04. | Introduction to Internet of Things: (<i>Coursera, University of San Diego</i>): Teaches about the basics of Internet of Things and how we have arrived at the current technological standpoint in this field and demonstrates the use of Dragon board for various projects. [VIEW] | <i>Oct '24</i> |

EXTRA-CURRICULAR

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| 01. | Graphics Lead: Iris The Photographic Society of UEMK | <i>Sep '24 - Dec '24</i> |
| 02. | Media Lead: Iris The Photographic Society of UEMK | <i>Dec '24 - Aug '25</i> |
| 03. | Tech Lead: Iris The Photographic Society of UEMK | <i>Dec '24 - Aug '25</i> |
| 04. | Graphics Lead: Computer Society of India UEM Student Chapter | <i>Mar '25 - Jan '26</i> |
| 05. | Manager: Computer Society of India UEM Student Chapter | <i>Aug '25 - Jan '26</i> |
| 06. | Vice President: Computer Society of India UEM Student Chapter | <i>Jan '26 - Present</i> |
| 07. | Class Representative | <i>Jan '26 - Present</i> |
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